

MOHAN RAJ SENTHILNATHAN

PRINCIPAL SOFTWARE ENGINEER
DISTRIBUTED SYSTEMS | AI AGENTS

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PROFESSIONAL OVERVIEW

Principal Software Engineer with 12+ years of experience building scalable, secure distributed systems and cloud-native platforms. Proven expertise in modernizing legacy architecture and developing AI agents using the Claude Agent SDK to automate workflows, reduce manual effort, and improve system efficiency.

WORK EXPERIENCE

Gen (Formerly NortonLifeLock / Symantec) - Chennai Jan 2017 – Present

Principal Software Engineer (Jul 2023) | Senior SWE (Aug 2020) | SWE (Jan 2017)

Multi-Tenant Web Portal:

- Architected and delivered a cloud-native multi-tenant platform using .NET Core, React, Azure, Redis, and Cosmos DB; scaled across Norton, Avast, and Avira **serving 400K daily active users**.
- Built AI-powered test triage (Claude Agent SDK) and automated defect remediation framework resolving up to 60% of regression failures without human intervention.
- Engineered an AWS Bedrock tool to summarize discussions and auto-generate Jira stories and defects from test results.
- Architected a security layer using OpenID Connect and RESTful APIs for distributed multi-tenant auth.
- Optimized Cosmos DB queries (selective fields, custom indexing) and integrated Redis cache for cross-server sync.

Email Guardian:

- Designed Feedback Loop enhancements with automatic re-scans via Pub/Sub; built a high-throughput Metrics API using Kafka and GCP BigQuery for real-time analytics.
- Implemented Database Sharding for horizontal scalability, enabling Norton tenant onboarding.
- Migrated Nginx to free tier via custom open source auth modules, reducing licensing costs.
- Developed a microservice orchestrator using RabbitMQ for scalable message-driven workflows, processing ~43M emails scanned per day.

Leadership:

- Mentored a team of 15 engineers, led code reviews, and championed knowledge-sharing sessions and design discussions. Led AI tool adoption including prompt engineering and spec design across the team.

HCL Technologies (Client: Cisco Systems) - Chennai Mar 2015 – Dec 2016

Member Technical Staff

- Developed REST APIs and backend services for Cisco Identity Services Engine (ISE).

Amazon Development Center — Chennai

Oct 2013 – Mar 2015

Device Associate

- Owned module-level testing for Amazon Fire Phone & Kindle devices, automating key test cases to enhance release *quality*.

SKILLS

Languages: C#, Java, Python, JavaScript, Typescript

AI & Agents: Claude Agent SDK, AWS Bedrock, LangChain, RAG pipelines

Frameworks: .NET Core, Django, React, Angular, Node.js

Cloud & DevOps: Azure, GCP, AWS, Kubernetes, Docker, Redis, Nginx, TeamCity, Azure DevOps, Git

Data: PostgreSQL, Cosmos DB, Kafka, RabbitMQ

Observability: Prometheus, Grafana, Datadog, Splunk

Architecture: Microservices, Event-Driven, CQRS, Domain-Driven Design, SOLID

ACHIEVEMENTS

Patent

Systems and Methods for Restricting Capture of Self-Portrait Images on Mobile Devices
Patent No: 11528355
Issued: Dec 13, 2022

EDUCATION

B. Tech in Information Technology

Anna University
2013

PROFESSIONAL DEVELOPMENT

Certification

Microsoft Certified: Azure Fundamentals (AZ-900)
2022